Norms and Low Rank Approximation

Math 5250 — Week 7

Overview

- Norms on vector spaces; all norms on a finite-dimensional space are equivalent.
- Matrix norms: the operator (spectral) norm and the Frobenius norm, and their relation to singular values.
- Proof that the truncated SVD A_k is the best rank k approximation in both norms.
- The Courant minimax principle.

Reference: Conrad, Computing the Norm of a Matrix.

Vector Space Norms

Norms

Let V be a vector space over $F = \mathbf{R}$ or $\mathbf{C}$. A **norm** on V is a function $\|\cdot\| : V \to \mathbf{R}$ such that:

- $\|v\| \geq 0$, with equality if and only if $v = 0$;
- $\|v + w\| \leq \|v\| + \|w\|$;
- $\|cv\| = |c|\|v\|$ for $c \in F$.

Dropping "$\|v\| = 0$ only if $v = 0$" gives a **seminorm**.

A norm makes V a metric space with $d(v, w) = \|v - w\|$.

Two Key Examples

The sup norm relative to a basis $e_1, \ldots, e_n$:

$$\left\| \sum_{i=1}^{n} a_i e_i \right\|_{\sup} = \max\{|a_1|, \ldots, |a_n|\}.$$

The ℓ^2 (Euclidean) norm from an inner product:

$$\|v\| = \sqrt{\langle v, v \rangle}.$$

For the usual dot product, this gives Euclidean distance.

The Triangle Inequality for ℓ^2

Cauchy–Schwarz: $|\langle v, w\rangle| \le \sqrt{\langle v, v\rangle\langle w, w\rangle}$. So

$$\begin{aligned}\|v + w\|^2 &= \langle v, v\rangle + 2\,\mathrm{Re}\langle v, w\rangle + \langle w, w\rangle \\ &\le \|v\|^2 + 2|\langle v, w\rangle| + \|w\|^2 \\ &\le \|v\|^2 + 2\|v\|\|w\| + \|w\|^2 = (\|v\| + \|w\|)^2.\end{aligned}$$

Take square roots.

Equivalent Norms

Definition. Norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on V are **equivalent** if there are constants $A, B > 0$ with

$$A\|v\|_a \leq \|v\|_b \leq B\|v\|_a \quad \text{for all } v \in V.$$

Proposition. All norms on a finite-dimensional vector space V over $\mathbf{R}$ or $\mathbf{C}$ are equivalent.

Strategy: fix a basis $e_1, \ldots, e_n$ and show that any norm ρ is equivalent to $\|\cdot\|_{\sup}$ for that basis.

The Upper Bound

For $v = \sum_{i=1}^{n} a_i e_i$:

$$\rho(v) \leq \sum_{i=1}^{n} |a_i| \rho(e_i) \leq \Big(\sum_{i=1}^{n} \rho(e_i) \Big) \|v\|_{\sup}.$$

So $\rho(v) \leq B\|v\|_{\sup}$ with $B = \sum_i \rho(e_i)$.

Lemma. ρ is continuous for the metric given by $\|\cdot\|_{\sup}$:

$$|\rho(x) - \rho(y)| \leq \rho(x - y) \leq B\|x - y\|_{\sup}.$$

This much works for seminorms too. The next step needs $\rho(v) = 0 \Rightarrow v = 0$.

Compactness of the Sphere

Lemma. The unit sphere $S = \{v : \|v\|_{\sup} = 1\}$ is compact.

Proof.

- The closed ball $\{\|v\|_{\sup} \leq 1\}$ is a finite product of intervals $[-1, 1]$ (or closed disks $|z| \leq 1$), so it is compact.
- The open ball $\{\|v\|_{\sup} < 1\}$ is a finite product of open intervals (or open disks), so it is open.
- S is the closed ball minus the open ball, so it is closed, hence compact. $\square$

The Lower Bound

ρ is continuous on the compact set S, so it attains a minimum A, and $A > 0$ since ρ is nonzero on S.

For $v \neq 0$:

$$\rho\Big(\frac{v}{\|v\|_{\sup}}\Big) \geq A \quad \Longrightarrow \quad \rho(v) \geq A\|v\|_{\sup}.$$

Altogether:

$$A\|v\|_{\sup} \leq \rho(v) \leq B\|v\|_{\sup}. \qquad \square$$

Remarks and Consequences

- The key input is compactness of closed intervals in $\mathbf{R}$ or closed disks in $\mathbf{C}$ (Heine–Borel).
- Conrad, Equivalence of Norms: all norms on a finite-dimensional vector space over a complete valued field are equivalent.

Corollary. For finite-dimensional V, W over $\mathbf{R}$ or $\mathbf{C}$:

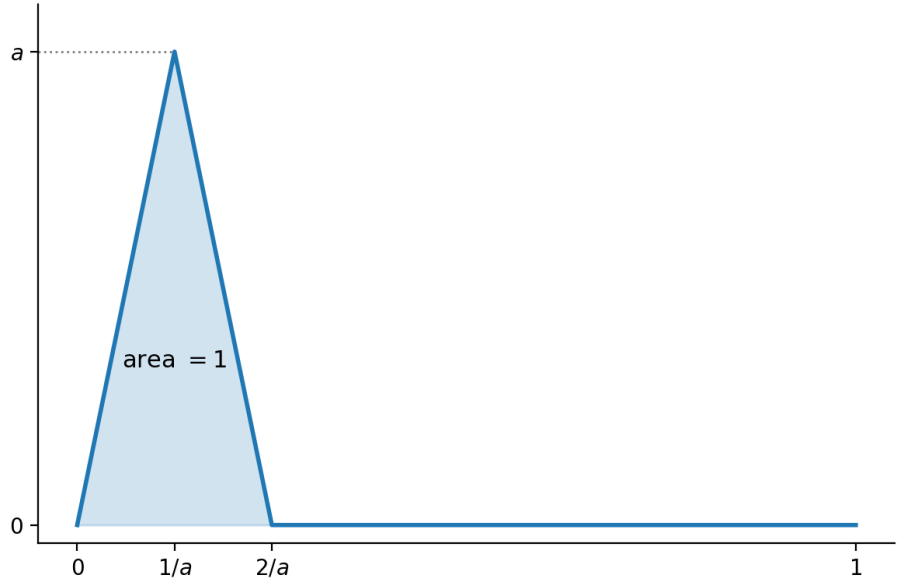
- every linear map $V \to W$ is continuous in any norm topology;
- bounded, open, closed and compact sets are the same for every norm.

Infinite Dimensions Are Different

On $C([0,1])$ consider

$$\|f\|_{\sup} = \sup_{0\le x\le 1} |f(x)|, \qquad \|f\|_{L^1} = \int_0^1 |f(x)|\,dx.$$

A spike of height a and area 1 has $\|f\|_{\sup} = a$ and $\|f\|_{L^1} = 1$. So there is no B with $\|f\|_{\sup} \le B\|f\|_{L^1}$: the norms are not equivalent.



Matrix Norms

Matrices as Vectors

$M_{n\times m}(F) \cong F^{nm}$, so matrices have:

- the sup norm $\|A\|_{\sup} = \max_{i,j} |a_{ij}|$;
- the **Frobenius norm** $\|\cdot\|_F$, the ℓ^2 norm on F^{nm}.

All norms on $M_{n\times m}(F)$ are equivalent.

The Seminorm ρ_A

For $A \in M_{n\times m}(F)$ let

$$\rho_A(v) = \|Av\|,$$

where $\|v\|$ is the ℓ^2 norm on F^m and $\|Av\|$ is the ℓ^2 norm on F^n.

ρ_A is a seminorm on F^m:

- $\rho_A(v) \geq 0$;
- $\rho_A(v+w) = \|Av + Aw\| \leq \rho_A(v) + \rho_A(w)$;
- $\rho_A(av) = |a|\rho_A(v)$.

So ρ_A is continuous, and $\rho_A(v) \leq C\|v\|$ for some $C \geq 0$.

The Operator Norm

Proposition. There is a smallest $b \geq 0$ with $\rho_A(v) \leq b\|v\|$ for all v.

Proof.

- The sphere $\{v \in F^m : \|v\| = 1\}$ is closed and bounded, hence compact, so ρ_A attains a maximum $b = \rho_A(v_0)$ on it.
- For $v \neq 0$, $\rho_A(v/\|v\|) \leq b$, so $\rho_A(v) \leq b\|v\|$.
- If $\rho_A(v) \leq c\|v\|$ for all v, then taking $v = v_0$ gives $b \leq c$. $\square$

Definition. The **operator norm** of A is

$$\|A\| = \max_{v \in F^m,\ \|v\|=1} \|Av\|.$$

Properties of the Operator Norm

Proposition. The operator norm is a norm on $M_{n\times m}(F)$, and for $B \in M_{m\times p}(F)$:

- $\|AB\| \leq \|A\|\|B\|$;
- $\|A^*\| = \|A\|$;
- $\|A^*A\| = \|AA^*\| = \|A\|^2$;
- $|\langle Av, w\rangle| \leq \|A\|\|v\|\|w\|$ for $v \in F^m$, $w \in F^n$;
- $\|A\|_{\sup} \leq \|A\| \leq \sqrt{nm}\,\|A\|_{\sup}$.

So the operator norm gives $M_{n\times m}(F)$ the topology of F^{nm}, and $M_{n\times m}(F)$ is complete for it.

(Conrad, Theorem 3.4.)

Proofs: Products and Inner Products

Products.

$$\|ABv\| \le \|A\|\|Bv\| \le \|A\|\|B\|\|v\|,$$

and $\|AB\|$ is the smallest such constant, so $\|AB\| \le \|A\|\|B\|$.

Inner products. By Cauchy–Schwarz,

$$|\langle Av, w\rangle| \le \|Av\|\|w\| \le \|A\|\|v\|\|w\|.$$

Proof: the Adjoint

A^* is the conjugate transpose ($A^\top$ when $F = \mathbf{R}$). By Cauchy–Schwarz,

$$\|Av\|^2 = \langle v, A^*Av\rangle \leq \|v\|\|A^*Av\| \leq \|A^*A\|\|v\|^2.$$

Maximize over unit v and use the product bound:

$$\|A\|^2 \leq \|A^*A\| \leq \|A^*\|\|A\|.$$

- If $A \neq 0$, divide by $\|A\|$: $\|A\| \leq \|A^*\|$.
- Apply this to A^*: $\|A^*\| \leq \|A\|$. So they are equal.
- Then $\|A\|^2 \leq \|A^*A\| \leq \|A\|^2$.
- Applying this to A^* gives $\|AA^*\| = \|A^*\|^2 = \|A\|^2$. $\square$

Self-Adjoint Matrices

Proposition. If $A \in M_n(F)$ and $A = A^*$, then

$$\|A\| = \max_{\|v\|=1} |\langle Av, v\rangle| = \max\{|\lambda| : \lambda \text{ an eigenvalue of } A\}.$$

The **Rayleigh quotient**

$$R(v) = \frac{|\langle Av, v\rangle|}{\|v\|^2}, \qquad v \neq 0,$$

is continuous and scale invariant, so it attains a maximum μ on the unit sphere and

$$|\langle Av, v\rangle| \leq \mu \|v\|^2 \quad \text{for all } v.$$

Since $|\langle Av, v\rangle| \leq \|A\|\|v\|^2$, we have $\mu \leq \|A\|$.

Proof: $\|A\| \le \mu$

Polarization. Since $A = A^*$,

$$\langle A(v+w), v+w\rangle - \langle A(v-w), v-w\rangle = 4\,\mathrm{Re}\langle Av, w\rangle.$$

So, using $\|v+w\|^2 + \|v-w\|^2 = 2(\|v\|^2 + \|w\|^2)$,

$$4|\,\mathrm{Re}\langle Av, w\rangle| \le \mu(\|v+w\|^2 + \|v-w\|^2) = 2\mu(\|v\|^2 + \|w\|^2).$$

For unit v, w this gives $|\,\mathrm{Re}\langle Av, w\rangle| \le \mu$; by scaling,

$$|\,\mathrm{Re}\langle Av, w\rangle| \le \mu\|v\|\|w\|.$$

Take $w = Av$: $\|Av\|^2 \le \mu\|v\|\|Av\|$, so $\|Av\| \le \mu\|v\|$. Hence $\|A\| = \mu$.

Proof: the Eigenvalue

Choose a unit vector u with $|\langle Au, u\rangle| = \|A\|$, and let $\alpha = \langle Au, u\rangle = \pm\|A\|$ (real since $A = A^*$). Then

$$\begin{aligned} 0 \le \|Au - \alpha u\|^2 &= \|Au\|^2 - 2\alpha\langle Au, u\rangle + \alpha^2 \\ &= \|Au\|^2 - \alpha^2 \le \|A\|^2 - \alpha^2 = 0. \end{aligned}$$

So $Au = \alpha u$, and $\pm\|A\|$ is an eigenvalue.

Every eigenvalue satisfies $|\lambda|\|v\| = \|Av\| \le \|A\|\|v\|$, so

$$\|A\| = \max_i |\lambda_i|. \qquad \square$$

The Spectral Norm

Corollary. For any $n \times m$ matrix A, $\|A\| = \sigma_1$, the largest singular value of A.

Proof. $\|A\|^2 = \|A^*A\|$, and A^*A is self-adjoint with eigenvalues σ_i^2 (and zeros). The largest is σ_1^2. $\square$

This is why the operator norm is also called the **spectral norm**.

The Frobenius Norm

$$\|A\|_F^2 = \sum_{i,j} |a_{ij}|^2.$$

Lemma. The inner product $\langle (a_{ij}), (b_{ij}) \rangle = \sum a_{ij}\overline{b_{ij}}$ on $M_{n\times m}(F)$ is

$$\langle A, B \rangle = \operatorname{trace}(B^* A), \qquad \text{so} \qquad \|A\|_F^2 = \operatorname{trace}(A^* A).$$

Proposition. If $\sigma_1 \geq \cdots \geq \sigma_r$ are the singular values of A, then

$$\|A\|_F = \sqrt{\sum_{i=1}^{r} \sigma_i^2}.$$

Proof. $\operatorname{trace}(A^*A)$ is the sum of the eigenvalues of A^*A, which are the σ_i^2 and zeros. $\square$

The spectral norm sees the largest singular value; the Frobenius norm sees them all.

Low Rank Approximation: Proofs

The Problem

Given $A \in M_{n\times m}(F)$, find A_k of rank at most k with

- $\|A - A_k\| \leq \|A - B\|$ (operator norm), and
- $\|A - A_k\|_F \leq \|A - B\|_F$ (Frobenius norm)

for all B of rank at most k.

The answer to both is the truncated SVD. (Following Wikipedia, Low-rank approximation.)

A and A^* have the same rank and norms, so assume $n \leq m$, and let

$$A = U\Sigma V^*, \qquad \sigma_1 \geq \cdots \geq \sigma_n \geq 0.$$

Two Lemmas

Let $v_1, \ldots, v_m$ be the columns of V (the right singular vectors).

Lemma A. For $1 \leq j \leq n$, if $w = \sum_{i=1}^{j} \gamma_i v_i$ is a unit vector, then $\|Aw\| \geq \sigma_j$.

Proof. $\sum_{i=1}^{j} |\gamma_i|^2 = 1$. $V^* w$ has entries $\gamma_1, \ldots, \gamma_j, 0, \ldots, 0$, so $h = \Sigma V^* w$ has entries $\sigma_1\gamma_1, \ldots, \sigma_j\gamma_j, 0, \ldots, 0$. Since U is unitary,

$$\|Aw\|^2 = \|Uh\|^2 = \sum_{i=1}^{j} \sigma_i^2 |\gamma_i|^2 \geq \sigma_j^2 \sum_{i=1}^{j} |\gamma_i|^2 = \sigma_j^2. \qquad \square$$

Lemma B. Set $\sigma_i = 0$ for $n < i \leq m$. If w is a unit vector in the span of $v_j, \ldots, v_m$, then $\|Aw\| \leq \sigma_j$.

Proof. The same computation. $\square$

The Operator Norm Result

Proposition. For $k < n$ let $A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^*$. Then for every B of rank at most k,

$$\|A - B\| \geq \|A - A_k\| = \sigma_{k+1}.$$

Proof.

- $\|A - A_k\| = \sigma_{k+1}$, the largest singular value of $A - A_k = \sum_{i>k} \sigma_i u_i v_i^*$.
- Let $S = \operatorname{span}(v_1, \ldots, v_{k+1})$, of dimension $k+1$. Since $B(S)$ has dimension at most k, rank–nullity gives a unit vector $w \in S \cap \ker B$.
- Then, by Lemma A with $j = k+1$,

$$\|A - B\|^2 \geq \|(A - B)w\|^2 = \|Aw\|^2 \geq \sigma_{k+1}^2. \qquad \square$$

The Frobenius Norm: Setup

Since the $u_i v_i^*$ are orthonormal in the Frobenius inner product,

$$\|A - A_k\|_F^2 = \Big\| \sum_{i=k+1}^{n} \sigma_i u_i v_i^* \Big\|_F^2 = \sum_{i=k+1}^{n} \sigma_i^2.$$

We must show $\|A - A_k\|_F^2 \leq \|A - B\|_F^2$ for all B of rank at most k.

Lemma (Weyl). If $A = A' + A''$ and $i + j - 1 \leq n$, then

$$\sigma_{i+j-1}(A) \leq \sigma_i(A') + \sigma_j(A'').$$

Proof of Weyl's Inequality

Note $\sigma_1(A - A_i) = \sigma_{i+1}(A)$, with $A_0 = 0$. So, by the triangle inequality for the spectral norm,

$$\begin{aligned}\sigma_i(A') + \sigma_j(A'') &= \sigma_1(A' - A'_{i-1}) + \sigma_1(A'' - A''_{j-1}) \\ &\geq \sigma_1(A - A'_{i-1} - A''_{j-1}).\end{aligned}$$

- $A'_{i-1} + A''_{j-1}$ has rank at most $i + j - 2$: the column space of $R + S$ lies in the sum of the column spaces of R and S.
- By the operator norm result with $k = i + j - 2$,

$$\sigma_1(A - A'_{i-1} - A''_{j-1}) \geq \sigma_{i+j-1}(A). \qquad \square$$

The Frobenius Norm Result

Let B have rank at most k. Apply Weyl to $A = (A - B) + B$ with $1 \leq i \leq n - k$ and $j = k + 1$:

$$\sigma_i(A - B) + \sigma_{k+1}(B) \geq \sigma_{i+k}(A).$$

But $\sigma_{k+1}(B) = 0$, so $\sigma_i(A - B) \geq \sigma_{i+k}(A)$, and

$$\begin{aligned} \|A - B\|_F^2 &= \sum_{i=1}^{n} \sigma_i(A - B)^2 \geq \sum_{i=1}^{n-k} \sigma_{i+k}(A)^2 \\ &= \sum_{i=k+1}^{n} \sigma_i(A)^2 = \|A - A_k\|_F^2. \qquad \square \end{aligned}$$

The Courant Minimax Principle

Minimax for Singular Values

Proposition. For $1 \leq k \leq n$, with S, T ranging over subspaces of F^m,

$$\sigma_k(A) = \max_{\dim S = k} \min_{u \in S,\ \|u\|=1} \|Au\| = \min_{\dim T = m-k+1} \max_{u \in T,\ \|u\|=1} \|Au\|.$$

Proof of the Minimax Formulas

$\geq$: Take $S = \text{span}(v_1, \ldots, v_k)$. By Lemma A,

$$\min_{u \in S,\ \|u\|=1} \|Au\| \geq \sigma_k.$$

$\leq$: Let S be any subspace of dimension k, and $T = \text{span}(v_k, \ldots, v_m)$, of dimension $m - k + 1$.

- $\dim S + \dim T > m$, so $S \cap T$ contains a unit vector w.
- By Lemma B, $\|Aw\| \leq \sigma_k$, so $\min_{u \in S, \|u\|=1} \|Au\| \leq \sigma_k$.

Second equality: similarly, $T = \text{span}(v_k, \ldots, v_m)$ gives $\leq$ by Lemma B, and any T of dimension $m - k + 1$ meets $\text{span}(v_1, \ldots, v_k)$, giving $\geq$ by Lemma A. $\square$

Courant's Principle

Strictly speaking, the **Courant minimax principle** says: if M is an $m \times m$ Hermitian matrix with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_m$, then

$$\begin{aligned} \lambda_k &= \max_{\dim S = k} \min_{x \in S,\ \|x\|=1} \langle Mx, x \rangle \\ &= \min_{\dim T = m-k+1} \max_{x \in T,\ \|x\|=1} \langle Mx, x \rangle. \end{aligned}$$

With $M = A^* A$, the eigenvalues are σ_i^2 and

$$\langle Mx, x \rangle = \langle Ax, Ax \rangle = \|Ax\|^2,$$

so our result is a special case.